Cost-Effective Online Uncertainty Quantification for LLM Agents via a Single Characterized External Estimator

Anonymous authors
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Abstract

Every serious deployment of LLM agents needs the same missing signal: a per-step, online estimate of whether the agent's next action should be trusted. We show the field's default sources of that signal fail structurally. Reading the acting model's own uncertainty does not transfer: across 62,065 labelled steps from six model families, the best intrinsic signal changes with every (model, benchmark) pair, and evaluating at the trajectory level is confounded outright, since episode length alone predicts failure at AUROC near 1.0. We propose reading the step from outside instead: one frozen, training-free external estimator that scores any agent's steps from environment-grounded evidence in a single forward pass. Characterized once, it outperforms the best self-assessment of 9 of 11 agents on the strictest error definition and all 11 under every other, including agents it cannot introspect, and outperforms gpt-4o as a judge at roughly one hundredth the cost. Decomposing why, we find reader capability dominates, with evidence of a smaller average self-leniency effect: models tend to under-report their own errors, though not uniformly across agents. Most surprisingly, the estimator's decision threshold is itself a transferable property: a map from its unlabelled score statistics to the optimal operating point, learned once on reference agents, places 12 of 16 held-out cuts within tolerance across four estimator families, zero target labels. We release the corpus, whose three-construct error taxonomy shows that the verdict of any agent-UQ method depends on the definition of error grading it.

1 Introduction

An agent halfway through a task proposes its next action. Whether that step was right will be nearly self-evident once the episode ends; right now, with only the prefix, it is the one thing the deployment most needs to know and cannot see. Every mechanism that would make agents safe to deploy at scale (flagging, escalation, selective autonomy, auditing) presupposes this per-step, online risk estimate (Oh et al., 2026; Ren et al., 2023). The field's foundational treatment of agent uncertainty names its two blockers: no reliable estimator under agentic constraints, and almost no step-level ground truth to test one against (4 of 44 benchmarks) (Oh et al., 2026). This paper supplies both, and in doing so overturns three working assumptions (Fig. 1).

First: no tested intrinsic signal transfers. The standard approach elicits the acting model's own uncertainty. Across eleven complete self-assessment arms we find there is nothing stable to elicit: the best intrinsic metric flips per (model, benchmark), transferring a recipe costs up to 0.10 AUROC, and the models that most need monitoring are the worst at introspection. Worse, the granularity at which the field evaluates (the completed trajectory) is confounded: episode length alone separates success from failure at AUROC ≈ 1.0, so a trajectory-level score can look excellent while knowing nothing about steps (§5.1).

Second: the fix is an instrument, not a better metric. We decouple estimation from execution: one frozen open-weight LLM, prompted, never trained, reads an evidence bundle for each step of *any* agent and emits a risk score in one forward pass, prefill plus one token. This single characterized estimator beats the best self-assessment of every agent below or at its own capability tier under every definition of step error we test (9 of 11 arms on the strictest construct, with the two exceptions being the most capable agents reading themselves), reads closed API agents it cannot introspect better than they read themselves, and beats gpt-4o run as a judge on every error construct at $\sim$1/100th the measured per-step cost (§5.2–5.3). A matched-capability decomposition shows reader capability dominates what the external position buys, with evidence of a smaller average self-leniency effect: models tend to score their own mistakes leniently (§5.2). The estimator's evidence law is measured, not assumed: an action alone is chance; trajectory history carries the signal; the environment's own replies are the best evidence available online.

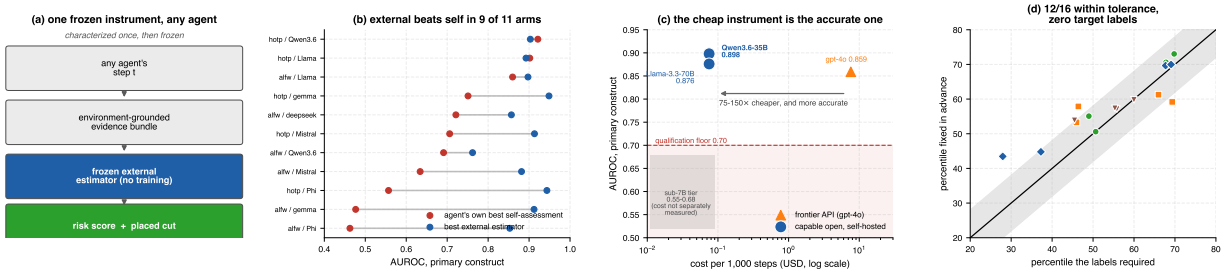


Figure 1: The paper in one figure. (a) One frozen external estimator reads environment-grounded evidence for each step of any agent and emits a risk score. (b) Best external reading versus each agent's best self-assessment, all eleven arms. (c) Cost versus accuracy per estimator tier, including the frontier judge. (d) Held-out threshold transfer: 12 of 16 cuts inside tolerance, zero target labels.

Third: even the threshold is a property of the instrument. Prior uncertainty-aware agents calibrate a decision threshold per task from labels and stop (Han et al., 2024). We show the threshold itself transfers: the optimal cut percentile is a learnable function of the estimator's unlabelled score statistics, fitted once on reference agents. On held-out agents, with placements fixed before any of their labels were read, it lands 12 of 16 cuts inside an 8-point tolerance across four estimator families, zero target labels, and beats the estimator's own binary judgment almost everywhere; when target labels exist, roughly 500 labelled steps suffice, and free environment violation signals reach the same accuracy on day one (§5.6). The failures are as informative as the successes: no fit-quality diagnostic we test, including one frozen in advance for a prospective test, predicts which cuts miss; what such statistics measure is how well the map is determined, not how wrong its placement will be.

What counts as an error is itself a result. Rebuilding all 62k step labels under three disclosed constructs (violation, outcome, judgment) moves 161 of 240 evaluation cells and can flip a method's verdict outright: agent-UQ results are construct-relative, and papers that leave the construct implicit are reporting a number whose meaning they have not fixed (§3).

Contributions. (1) Impossibility: no intrinsic UQ signal transfers across agents, and trajectory-level evaluation is length-confounded. (2) A single characterized external estimator as a universal step-level instrument within an environment family: independence decomposed into capability and measured self-leniency; evidence law; score-over-verdict; tier and frontier economics. (3) The operating-point map: threshold transfer with zero target labels, validated on held-out agents across four estimator families, with its failure modes mapped. (4) The corpus: 62k step-labels under three audited constructs with per-step provenance, and the demonstration that the construct choice changes conclusions.

2 Related Work

Foundations of agent UQ. Oh et al. (2026) formalize agent problem-solving as a trajectory of actions, observations, and environment states, define turn-level uncertainty $U(A_i \mid E_{i-1}, O_{i-1})$, identify four challenges (estimator selection, heterogeneous entities, uncertainty dynamics, benchmark scarcity), and find standard estimators near chance on τ^2-bench. Our estimator targets exactly their action-conditional term, externally; our results answer their first and fourth challenges by measurement, and where they suggest an auxiliary LLM for content the agent did not generate, ours is an adjacent instantiation that conditions on realized observations as evidence rather than modeling observation uncertainty.

Intrinsic and single-turn UQ. Token- and sequence-level signals, verbalized confidence, and prompted self-knowledge form the standard toolkit (Kadavath et al., 2022; Fadeeva et al., 2023); Tian et al. (2023) show RLHF models verbalize usefully calibrated confidence in single-turn QA, and Zhang and Zhang (2025) improve response-wise estimates by exploiting chain-of-thought. Our measurements scope rather than contradict these: in-generation verbalized confidence inverts for capable models during relational, stateful action selection, so setting and tier both matter. Kapoor et al. (2024) teach uncertainty by fine-tuning and observe that one strong model can estimate other models' uncertainty, including closed models, given a small labelled set; our independence result is the agentic, training-free, step-level extension, and our few-hundred-labelled-step calibration tier is their small-labelled-set clause, measured.

Agentic UQ methods. Existing methods either emit verdicts for completed or embedded trajectories (UALA's uncertainty-gated tool use with quantile thresholds over a labelled calibration set (Han et al., 2024); situation-weighted propagation (Zhao et al., 2024); multi-step propagation (Duan et al., 2025); tensor decompositions for multi-agent systems (Chen et al., 2026); verbalized-uncertainty control signals (Zhang et al., 2026a,c)) or assess steps by self-evaluation (stepwise self-confidence (Mavi et al., 2025); confidence elicitation for embodied action (Yu et al., 2025); clarification triggers (Matsnev, 2026)). Conformal uncertainty alignment lets planners ask for help (Ren et al., 2023); supervised predictors learn when to escalate (Bachar et al., 2026); CEB estimates pre-execution step confidence via an experience bank (Zeng et al., 2026), whose problem statement we adopt with a training-free estimator; ReDAct defers from a small actor to a large one on calibrated uncertainty (Piatrashyn et al., 2026). Against this line our estimator is step-granular, online, and external; our length-confound result cautions the trajectory-level evaluation regime; and our matched-capability decomposition gives the self-vs-external question its first controlled answer.

LLM-as-judge and verification. MT-Bench established judge promise and biases, including self-enhancement (Zheng et al., 2023); we quantify its step-level form (self-leniency) and design around it, and our annotation validation follows its agreement recipe run against environment-certain ground truth. Guideline-grounded verification (Zhang et al., 2026b) occupies one cell of our evidence grid, and our decoy-controlled measurement of rule-manufactured false positives is a caution such verifiers inherit. Step labels connect to process supervision (Lightman et al., 2024), whose aggregation choices are special cases of trajectory UQ.

Label shift. Our operating-point transfer lives in the prior-probability-shift setting of invariant class-conditionals with a moving mixing weight (Saerens et al., 2002; Lipton et al., 2018); we measure where that invariance actually holds for estimator score distributions, which we believe is new.

3 Problem Setup: The Temporal Gap, Step-Error Constructs, and Corpus

An agent searches a plausible entity; the search returns nothing. Failure, or a good decision unluckily resolved? The answer depends on the label construct, and we make that dependence explicit, measured, and released. Step labels come in three families (Table 1): **violation** labels, meaning inadmissible actions, exact state–action repeats, no-ops, malformed calls: wrong under every construct (the legal-command list was in the agent's prompt; a repeated pair has zero information gain) and certain by environment mechanics; **outcome** labels, meaning zero-result searches, gold-supporting-fact retrieval, expert-plan distance: hindsight-anchored, encoding oracle knowledge the agent lacks; and **judgment** labels: expert-style ex-ante annotations of decision quality, targeting the same construct as the paper, produced by an automated annotation pipeline (three disjoint models, none ever evaluated as an estimator: grok-4.3, DeepSeek-V4-Pro, gpt-5.6-sol). The primary evaluation construct is **violation+judgment**: a step is labelled incorrect iff it triggers a violation rule *or* the annotation marks it incorrect ($y = y_{\text{viol}} \vee y_{\text{judg}}$; violation steps are incorrect regardless of annotation), i.e., ex-ante decision quality with environment-certain violations folded in. Step-granular labels of this kind are the agentic analogue of process supervision (Lightman et al., 2024), released here with per-step provenance.

Automated annotation earns trust the way judge-based evaluation first did in QA, by measured agreement (Zheng et al., 2023), and we run that recipe against the environment: where mechanics give certain ground truth, the annotation errs 1.3%/0.8%/0.3–1.9% per violation sub-rule (2.0% overall excluding zero-result searches); the 9.1% headline is owned by the one sub-rule (31.5%) where the environment's verdict and ex-ante quality genuinely diverge, 2,233 unanimous miscalls, every one released. We accordingly treat these annotations as ground-truth ex-ante labels throughout, with the residual assumption confined to one limitation (§7).

Rebuilding every label from environment signals and recomputing all results leaves the paper's conclusions standing. The swap still moved 161 of 240 evaluation cells by more than 0.05 AUROC; the same placement rule can pass under one error definition and fail under another; and the independence gain of §5.2 is $+0.119$ larger under hindsight labels than under the primary (difference-of-differences, 9/11 arms, CI $[0.036, 0.205]$): hindsight labels measure partly different things, quantified. This is one concrete form of the survey's call for evaluation beyond task failure: *the verdict of a UQ method depends on which construct of "step error" grades it.*

Corpus. Two environments (ALFWorld (Shridhar et al., 2021), HotpotQA (Yang et al., 2018)), ReAct-style agents (Yao et al., 2023) spanning Phi-4-mini, gemma-3-4b, Mistral-7B, Qwen3.6-35B, Llama-3.3-70B, and an API-only model (ALFWorld only): 16 arms, 61 (estimator×target) cells, 1.22M scored lines, and a labelled file of 62,065 steps $\times$ 39 columns with per-step provenance: five label columns and both label passes (annotation labels and environment-derived labels; results use environment-derived labels unless an analysis is marked annotation-only). Five agent arms

Table 1: Three step-error constructs. The primary evaluation construct is violation+judgment (ex-ante decision quality).

construct	written by	certainty	audited error
violation	environment mechanics	absolute	n/a (certain by construction)
outcome	environment + oracle knowledge	high, hindsight	n/a (hindsight-defined)
judgment	annotation pipeline (disjoint)	ex-ante, audited	2.0% ex-A4 (certain band)

Figure 2: Corpus Architecture and Two-Pass Evaluation Pipeline

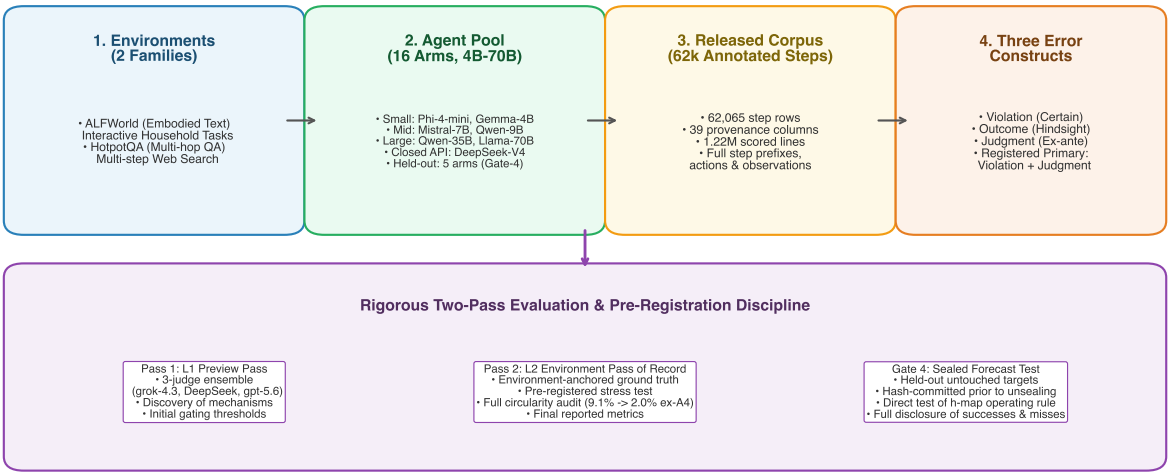


Figure 2: Corpus and pipeline: two environments, sixteen arms, 62,065 labelled steps under three error constructs, scored by every estimator, with a two-pass label validation.

were held out of every experiment and reserved for the generalization test of §5.6.

4 Method: A Characterized External Estimator

4.1 The estimation protocol

Protocol. For step t the estimator reads an evidence bundle (task, full interaction history, the agent's reasoning, the proposed action, and an optional rubric) and answers a fixed $P(\text{True})$-style query; the score $U \in [0, 1]$ is the renormalized probability mass on the "incorrect" token: one prefill plus one generated token, greedy. Measured throughput: 3,300 assessments / 5 min / A100 at online prompt lengths, KV-prefix-shared with the trajectory. Every headline number is a pre-execution reading: the estimator sees exactly what the agent saw, never the observation its action produced, checked mechanically across all 28,050 prompts (Table 8, appendix). The probe is the cost-collapsed descendant of single-answer grading (Zheng et al., 2023): the graded rubric becomes one logit read. **Evaluation.** Balanced accuracy (BA) for decisions at a cut; AUROC for ranking; paired, episode-clustered bootstrap CIs (2,000 draws). Throughout, "calibration" refers to placing the decision threshold (operating-point calibration), not to probabilistic calibration of the score itself. Raw scores are not calibrated probabilities; a monotone recalibration on the already-labelled episodes fixes this (ECE $0.109 \rightarrow 0.019$) without moving the operating point (Appendix D); *capture* = the fraction of the fully-supervised gain over the verdict baseline that a label-free rule recovers. The capable stratum is Llama-3.3-70B and Qwen3.6-35B as estimators. We emphasize effect sizes and replication over counts; adjusted summaries are in Appendix A.

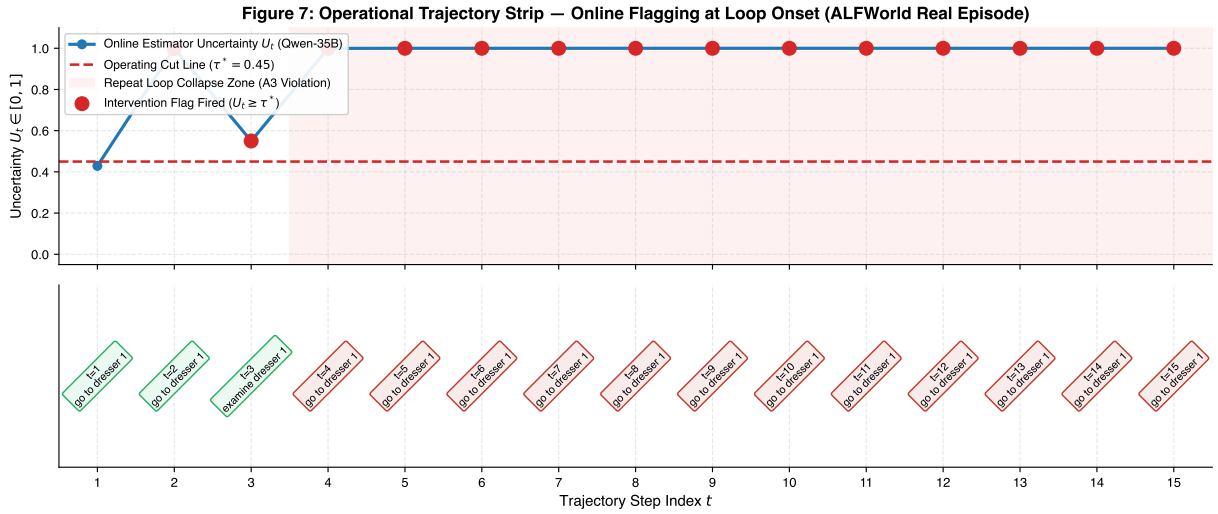


Figure 3: The instrument in operation on one real ALFWorld episode: per-step score U, the deployed cut, and flags firing at the onset of a repeat loop.

4.2 The operating-point map

Uncertainty-aware agents to date calibrate a decision threshold from a labelled set, per task, and stop; UALA's quantile over a calibration set is the canonical form (Han et al., 2024). We ask the question that practice leaves open: *is the threshold a property of the target (re-calibrate forever) or of the estimator (characterize once)?* This section is the paper's central mechanism contribution, and its answer is conditional in a way we can state exactly.

The mechanism. The optimal cut percentile is a learnable, approximately affine function of the estimator's own mean score on the target: $h : \bar{U} \mapsto \pi^*$, fitted once per (estimator, environment) on reference agents, leave-one-target-out. Deployment on a new agent consumes two unlabelled, streamable statistics of its own score stream, the mean (input to h) and the empirical quantiles (percentile $\to$ raw cut), giving two-layer, label-free adaptivity. Two measured facts make the map learnable with ordinary accuracy: the BA landscape is flat (median flat region 16 percentile points; ± 8 costs ≤ 0.01), and mean score tracks target error rate at $r = 0.960$–0.998 (Fig. 4). Evaluation, on reference and held-out agents alike, is deferred to §5.6.

5 Experiments

Six ways of reading a step and six ways of placing its threshold serve as baselines throughout this section (Fig. 5): for reading, the agent's own best intrinsic signal found by per-agent search, the weak-estimator tier, a supervised critic at a fixed annotation budget, a frontier API judge, and the capable external estimator's binary verdict, against its score channel; for placement, raw threshold transfer, a fixed percentile, an error-rate-indexed rule, an oracle handed the true error rate, and an unsupervised mixture cut, against the learned operating-point map. The verdict channel is compared in balanced accuracy in §5.3, its native currency, since a binary answer has no ranking to score.

5.1 No intrinsic signal transfers

Across eleven complete self-assessment arms covering the standard intrinsic toolkit (Kadavath et al., 2022; Fadeeva et al., 2023) (annotation labels; no post-hoc sign flips, sub-0.5 AUROCs reported as anti-signal), three distinct metrics win somewhere, and $P(\text{True})$ wins only 7/11; transferring a fixed recipe costs up to -0.10 AUROC. The trajectory level, where propagation-style methods report (Zhao et al., 2024; Duan et al., 2025), is confounded in cap-dominated settings, where failures run to the episode cap: length alone reaches AUROC ≈ 1.0 there (Fig. 6), and the error mass is agentic (loops, stagnation, caps: 40–87% of incorrect steps). In-generation verbalized confidence inverts for capable models (0.19–0.40) while remaining the best of a bad lot for small ones, consistent with pile-up degrading control (Han

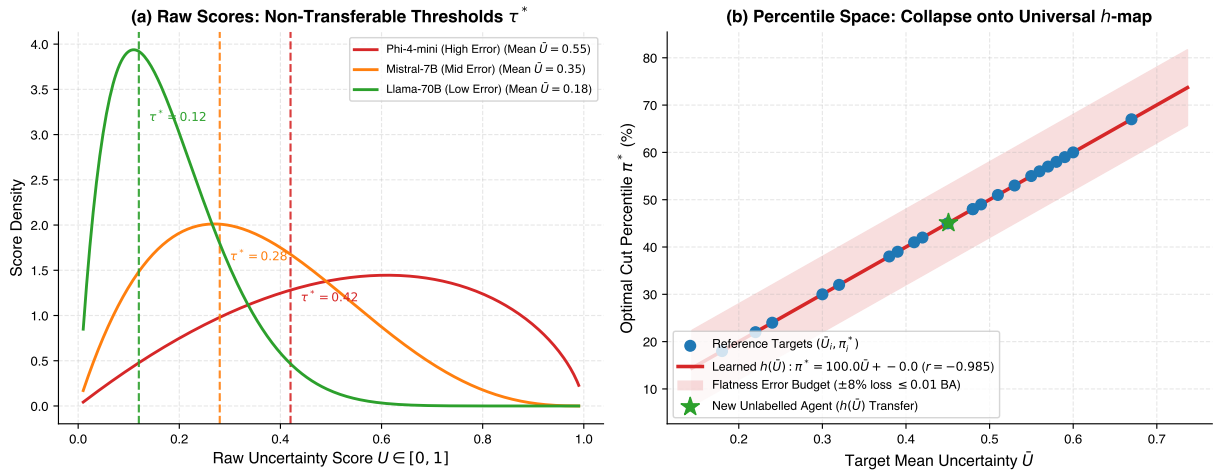


Figure 4: The operating-point map. Left: optimal raw cuts sit at different positions in each target's score distribution. Right: the same cuts collapse onto one affine line per (estimator, environment) in (mean score, optimal percentile) space; the shaded band is the ± 8-point flatness budget.

et al., 2024) and scoped against single-turn calibration (Tian et al., 2023) by setting and tier. Fragmentation means per-agent re-optimization forever, and never for closed agents; so we stop optimizing the agent side, and decouple.

5.2 Decoupling works: independence

Under environment-derived labels, the best external estimator beats the target's best self-assessment in 10 of 11 arms on the violation, outcome, judgment, and combined-environment constructs, and 9 of 11 on the primary, with CIs excluding zero in 7/8/10/7/10 arms per construct; restricted to capable estimators the pooled gain is $+0.222$ BA, CI $[0.180, 0.272]$, winning 11/11 countable external cells (Fig. 7). Margins on 4B-class targets under annotation labels were larger still ($+0.19$ to $+0.30$ AUROC; external readings 0.815–0.927 where self-best sat 0.62–0.73). The boundary is stated, not hidden: at the top of the capability range the two upward-probing cells are the exceptions (-0.0092, CI $[-0.0261, 0.0068]$; -0.0183): independence holds downward and peer-level.

What the external position buys, decomposed. The headline could be read two ways: externality matters, or a stronger model is simply a better reader. Both are true and separable. With the family confound removed the advantage is intact: a same-family estimator reading the held-out agents gains $+0.059$ AUROC over their own best self-assessment, 4/4 cells, every interval excluding zero. With the capability advantage removed it largely disappears (peer scale $+0.008$) and reverses among small models reading one another (-0.074): externality without capability costs accuracy (Appendix E). The residual, present even at peer scale as a directional asymmetry, is leniency: models under-read their own errors.

Self-leniency, measured three times. Why must the estimator be external? (i) Estimators score their own agents' incorrect steps with uncertainty depressed by $+0.128$ on average relative to cross-judging (correct steps $+0.048$). (ii) Feeding an estimator's self-scores into a competence map fitted on cross-judging misestimates the target's error rate by -0.140 and -0.105 where cross-cells average 0.052. (iii) The single unstable characterization fit of §5.6 had the estimator's own self-cell as its leverage point: the reason the forecast-tested variant excludes self-cells from its reference pools. This is self-enhancement bias (Zheng et al., 2023) quantified at step level, with its calibration consequences; the design excludes self-judging in the method *and* in the pools. The effect does not depend on the annotation: restricted to steps an environment rule marks incorrect, with no model-generated label anywhere in the loop, the same depression appears ($+0.102$ against $+0.114$ on the primary construct), the competence-map failure persists on ALFWorld (-0.121 against a 0.043 cross-cell baseline) though not on HotpotQA, and the unstable fit still has the estimator's own self-cell as its leverage point. Sign varies across arms (6 of 11), so we report the direction, not a pooled magnitude (Appendix E). The payoff of separation is standardization: heterogeneous agents, from 4B open weights to a closed API model, are read on one score scale by one instrument.

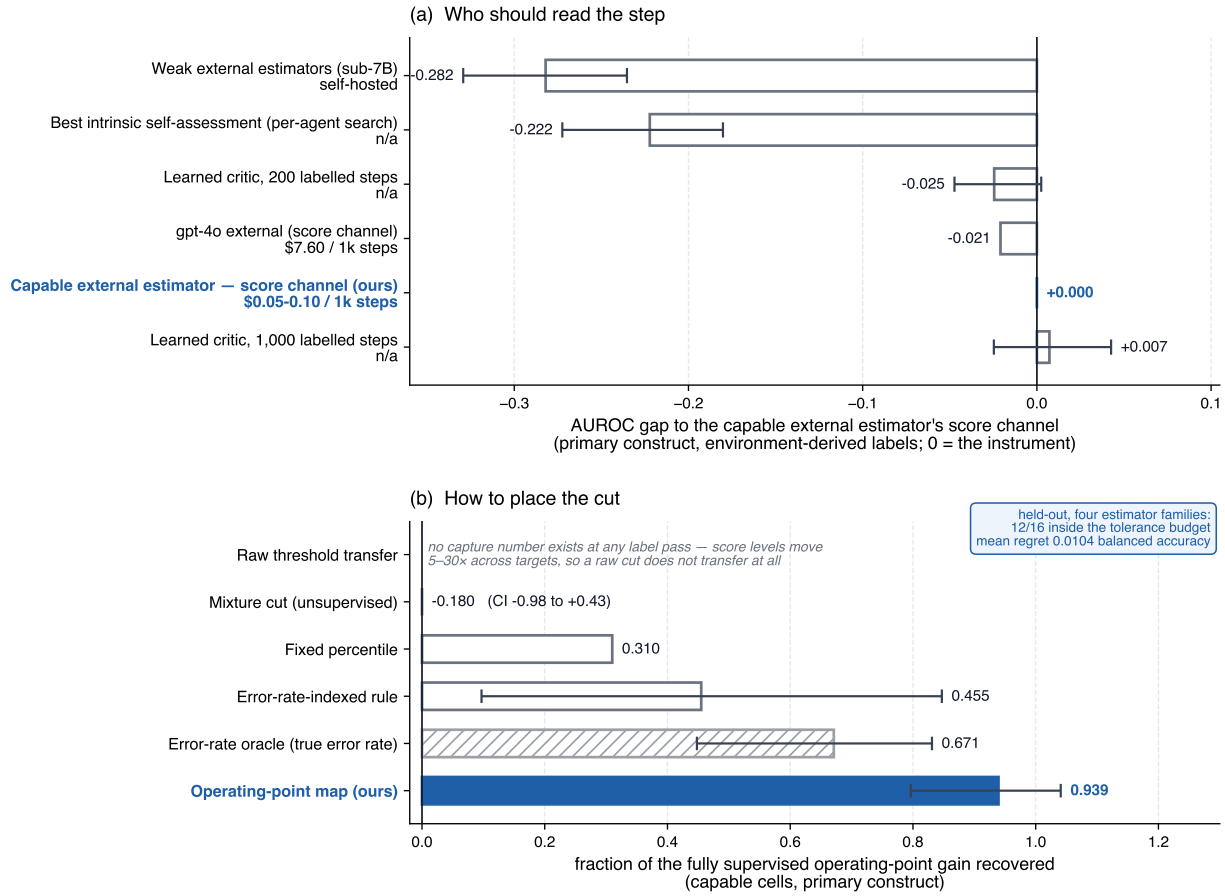


Figure 5: **Baselines. (a)** Step-level discrimination by estimator family on the primary construct under environment-derived labels, as the AUROC gap to the capable external estimator's score channel, with measured per-1k-step cost where applicable: the instrument dominates every alternative reading of the same steps, including each agent's own best signal found by per-agent search (-0.222), the weak-estimator tier (-0.282), a frontier API judge at 75–150$\times$ the cost (-0.021), and a supervised critic at the same annotation budget (-0.025 at 200 labelled steps, drawing level at 1,000). **(b)** Fraction of the fully supervised operating-point gain recovered by each placement scheme: the learned map (0.939) dominates every simpler transfer rule, including an oracle handed the true error rate (0.671; hatched, the one bar measured on the annotation-label pass).

5.3 What the estimator must read

The survey identifies three constraints any agentic estimator must clear: probability access, inference overhead, grounding. Each is resolved here by measurement.

Recipe invariance (the first characterization result. Under external estimation one probe recipe is stable across agents, environments, and estimators (within 0.02 of optimal in 133 of 145 configurations; Appendix A): the recipe instability of §5.1 is a property of self-assessment, not of the instrument, and characterization is therefore done once.

What it reads. A paired, decoy-controlled ablation (annotation labels, marked; $n \approx 300$/condition, 95% CIs; Table 2) gives the evidence law. An action alone is chance (0.512): step quality is relational, and QA judging where answer text suffices does not port. History is the signal carrier (0.697); the environment's realized response is the largest single increment ($+0.072$, to 0.813): the online estimator recovers hindsight evidence one step late, the temporal gap of §1 exploited rather than merely suffered. Given history and reasoning, the action token itself adds nothing (0.740 vs 0.741), and dropping the agent's reasoning costs -0.044 with overlapping CIs: black-box operation over the target is viable. A generic rubric recovers nearly all of a targeted rule's gain (0.804 vs 0.813) with no per-action typing; rules also *manufacture* false positives on rule-adjacent correct steps ($+0.155$ on legitimate revisits), so every

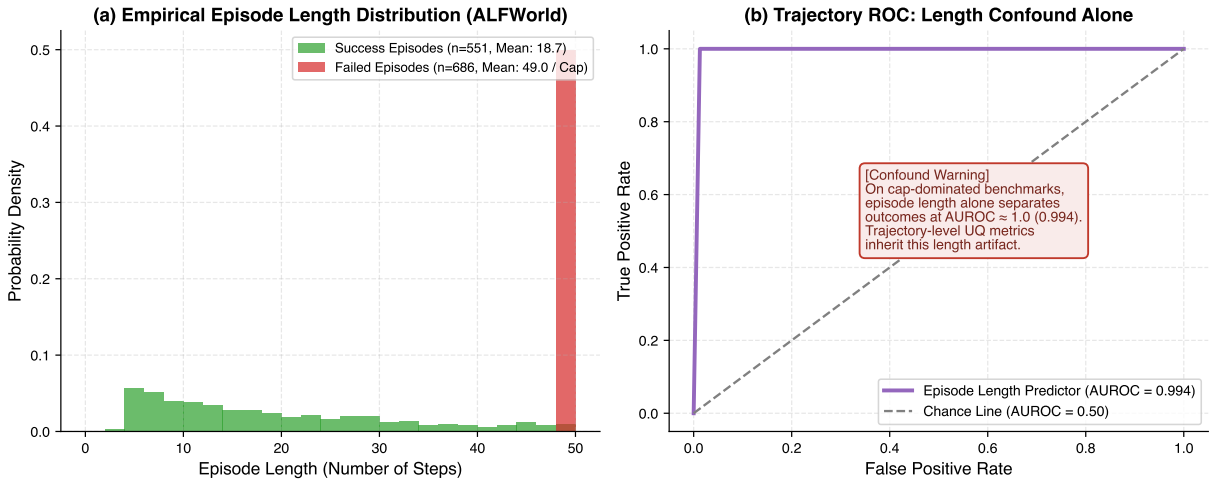


Figure 6: Trajectory-level evaluation is length-confounded: episode length alone separates success from failure at AUROC ≈ 1.0 on cap-dominated benchmarks.

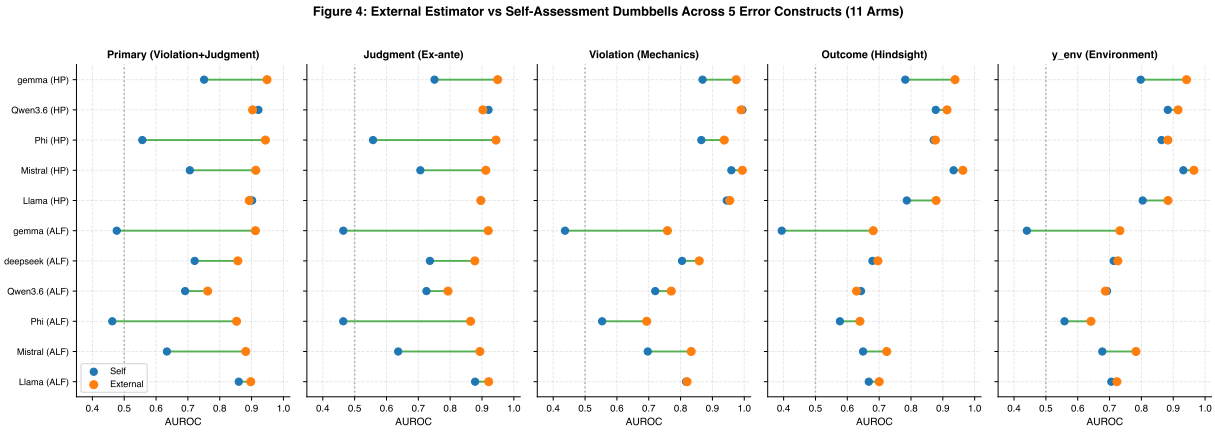


Figure 7: Independence across constructs: best self-assessment versus best external reading per arm, all five label constructs (environment-derived labels; paired episode-clustered CIs).

rule-augmented number is reported against a decoy floor (Zheng et al., 2023; Zhang et al., 2026b).

5.4 The probability, not the verdict

Which channel: the probability, not the verdict. The estimator's greedy binary verdict *is* its $P(\text{True})$ value at a fixed semantic cut (same prompt, same prefill, one logit apart) so channel choice is a fair, paired contest. Keeping the probability and fitting the cut out-of-sample recovers $+0.065$ BAover the verdict (CI $[0.052, 0.079]$; value $\geq$ verdict in 55/60 cells; fitted-cut optimism $+0.009$), and $+0.073$ to $+0.133$ per construct under environment-derived labels. The interesting part is *why* the verdict loses: its implicit threshold is genuinely adaptive (flag rate tracks the target's error rate at $r = +0.978/+0.956$ for capable estimators) and still misplaced: one under-flags low-error regimes (7% of steps flagged at a 15% error rate; BA 0.614), the other over-flags everywhere ($+0.121$). Adaptive is not optimal, and a bit has no knob; weak estimators' verdicts collapse outright while their scores stay recoverable via a cut.

Table 2: Evidence grid (Qwen3.6-35B estimator; $n \approx 300$/condition; annotation labels.

condition	contents	AUROC [95% CI]
C0	task + action	0.512 [0.438, 0.586]
C2	+ full history	0.697 [0.637, 0.755]
C3 (base)	+ reasoning	0.741 [0.675, 0.801]
C4	C3 − action	0.740 [0.673, 0.798]
C5	C3 + realized observation	0.813 [0.756, 0.865]
R1	C3 + generic rubric	0.804 [0.746, 0.853]
R2	C3 + targeted rule	0.813 [0.753, 0.866]
R3	C3 + decoy rule	0.728 [0.661, 0.789]

Table 3: Punchline: estimator tier × discrimination (mean external-cell AUROC, environment-derived labels, self-cells excluded) × measured cost. Qualification per §5.6.

estimator	viol+jud	judgment	outcome	$/1k steps	qualifies
Qwen3.6-35B	**0.898**	0.907	0.774	∼0.05–0.10	yes
Llama-3.3-70B	**0.876**	0.884	0.734	∼0.05–0.10	yes
gpt-4o	0.859	0.861	0.711	7.60	external only
Mistral-7B	0.679	0.682	0.759	∼0.05–0.10	no
gemma-3-4b	0.590	0.582	0.598	∼0.05–0.10	no
Phi-4-mini	0.546	0.546	0.744	∼0.05–0.10	no

5.5 Tier, frontier, and cost

What size, whose, at what cost. The capable-vs-below tier gap is $+0.282$ $[0.235, 0.329]$: a tier structure with a wide qualification gap, which is the refined answer to the original scaling hypothesis (no smooth plateau; a cliff, then diminishing differences). Within tiers, the 35B estimator's score matches or beats the 70B estimator's verdict on 11/11 shared targets with a few-hundred-labelled-step cut and 8/11 fully label-free (8/9 in cross-judging cells, the design's regime; both true losses are self-cells). One cell of color: on the API-only agent's own trajectories, the external mid-size read (0.800) beats the agent's self-reading at its own cut ($\sim$0.72–0.76). At the frontier, gpt-4o run as an external estimator (both channels, same framing verbatim, paired on identical steps) trails both capable open estimators on *every* construct (-0.021 primary, ahead in 8/22 cells), with a systematic asymmetry: its deficit is smallest on LLM-written judgment labels (-0.026) and roughly doubles on environment-anchored constructs (-0.037/-0.064/-0.061): same-species affinity between LLM judges and LLM-derived labels, while against the full open field it leads on judgment ($+0.147$) far more than on violation ($+0.047$). Adjusted-significance counts and framing-sensitivity checks are in Appendix A; the effect direction is consistent. Framing sensitivity is small (a decision-framed variant agrees with the trust framing on 91.4% of 740 shared steps, mean $|\Delta U| = 0.090$);. Two structural points complete the tier story: one frontier API in our pilot refuses logprobs outright (a frontier model can be the *agent* in this framework but not the *estimator*; knownness requires access) and the survey's hoped-for grounded verbalized channel is, on our measurements, the inverted one; the grounded alternative available today is the logprob channel on an estimator one hosts. Cost, measured: \$7.60 per 1k steps for the frontier arm (from the receipt) versus ∼\$0.05–0.10 self-hosted at market A100 rates, a configuration-dependent figure measured under our setup; at 1M steps/day that is ≈\$7,600 versus ∼8.5 A100-hours. For low-volume or infrastructure-free deployments the API path remains reasonable; in the continuous-monitoring regime the capable open estimator dominates on both axes (Table 3).

5.6 Validating the operating-point map

Baselines. We compare the map against every simpler transfer scheme a practitioner would try first. Raw score thresholds do not transfer: group score levels move 5–30× across targets. A fixed *percentile* transfers poorly (value $\geq$ verdict in only 55% of capable cells, capture 0.310). Indexing the percentile by the target's error rate embeds a misspecified law: even an *oracle* handed the true error rate places the cut suboptimally on half the cells where the indexed rule loses, because optimal percentiles deviate from $100(1-p)$ by up to ∼26 points. An unsupervised mixture cut underperforms throughout. A supervised critic trained on cheap trajectory features at the same annotation budget

does not match the training-free estimator: at 200 labelled steps it trails by 0.016–0.095 balanced accuracy across cells and at 1,000 it only draws level ($+0.001$, CI $[-0.031, +0.034]$) (excluding rule signals that define the primary label; appendix). Learning the percentile map directly dominates this family: leave-one-target-out, it recovers 94% of the fully supervised gain with zero target labels (capture 0.940 $[0.796, 1.041]$; over the verdict in 20/22 cells) and, notably, beats the error-rate oracle (capture 0.940 vs 0.67).

Generalization to held-out agents. Five agent arms were excluded from all development and reserved for a strict test: the map's predictions (cut percentile and resulting BA) were fixed before any label of these arms was examined. Reference pools exclude self-cells, per §5.2. On these unseen agents the map recovers 75% of the supervised ceiling and lands within the flatness budget in 6 of 8 evaluable cells, and the split is systematic rather than random: *characterization stability predicts transferability*. The stable characterization (Llama: jackknife swings 1.0–3.4 points, tight slope CIs) transfers with law-like regularity in our tests: 4/4 over the verdict, percentile misses of 0.0–6.1 points, including on the environment where the clean theory of §6 fails; the unstable one (Qwen: an 11.7-point swing, $5\times$ wider slope CI, identifiable in advance from the reference fit alone) misses by up to 11.5 points with two losses of 0.010 and 0.001; and the one agent larger than every reference agent misses by 31–55 points: the map interpolates, it does not extrapolate (Fig. 10, appendix). Two further estimators from families outside the first two (Gemma-3-27B, Mistral-Small-24B), run under the same fixed-before-evaluation protocol, extend the held-out record to 12 of 16 evaluable cells inside the budget across four estimator families (estimator-side diversity; target-family generalization remains open); both were the weakest qualifiers in the record and both beat their own binary answers in nearly every cell, with the misses concentrating where the reference agents' scores compress. Graded in accuracy rather than in percentile points, the held-out record is tighter than the tolerance criterion suggests: across all sixteen evaluable cells the transferred cut gives up 0.0104 balanced accuracy $[0.0077, 0.0132]$ against a cut fitted on the cell's own labels, and no cell gives up more than 0.0225. The four cells outside the ± 8-point budget give up 0.0095 $[0.0069, 0.0130]$, no more than the twelve inside it (0.0107 $[0.0072, 0.0143]$): the percentile criterion is a conservative proxy, and the misses it flags cost about one accuracy point, the same as the lands (Table in Appendix A).

5.7 Deployment: three label economies, one accuracy

In practice three routes set the operating point on a new agent, and they are statistically indistinguishable on the primary construct: one labelling pass over the deployment episodes (BA 0.808; the cut saturates by roughly 500 labelled steps, so an afternoon suffices — Table 5), free environment violation signals on day one (0.801, with a disclosed collapse to 0.672 on hindsight constructs), and the label-free map of §4.2 (0.799) where it applies. The choice is a label economy, not an accuracy ladder. All routes run online: a streaming replay tracks the batch rule to ≤ 0.001 including warm-up, and a supervised critic given the same 200 labelled steps trails every route by 0.016–0.095 BA, drawing level only at $1{,}000$ labels (excluding rule signals that define the primary label; appendix). The same operating points read as a selective-autonomy contract: at the deployed cut about 43% of steps run unsupervised at an error rate of 0.27 against a base rate of 0.57, so the instrument roughly halves the error rate on the traffic it admits (Fig. 13), and all three label economies land at the same point on that curve (within 0.012 coverage, 0.006 selective risk). Downstream, the per-step risk score is the primitive that deferral and escalation policies (Piatrashyn et al., 2026; Bachar et al., 2026) consume (Fig. 8; tables in appendix).

6 Analysis

Mechanism at honest size. The clean theory, in which class-conditional score distributions are invariant across targets and only the error rate moves (label shift; Lipton et al., 2018; Saerens et al., 2002), holds on one environment (within/between-class Wasserstein ratios 0.15–0.16) and *fails on the other* ($0.364/0.809$) where h nevertheless works: conditional invariance is sufficient, not necessary, and the surviving mechanism is the flat landscape plus a coarse affine regularity whose fitted slopes cluster (-0.49 to -0.67) across all eight (estimator, environment) fits; the environment split itself replicates on both later estimators (invariance holding on HotpotQA and failing on ALFWorld for every estimator tested). One further measurement we report without claiming: h applied *across* environments shows no measurable degradation (mean -0.0024 BA, CI $[-0.0047, +0.0001]$) on the same test where raw cuts lose 0.035–0.103: with $n{=}2$ environments this is an open question, stated as one (Figs. 11 and 12, appendix).

Qualification. The held-out results convert "does the map work?" into "for which estimators?", and we propose criteria accordingly. *Discrimination* (external-judging eligibility): primary-construct external AUROC ≥ 0.70: the six

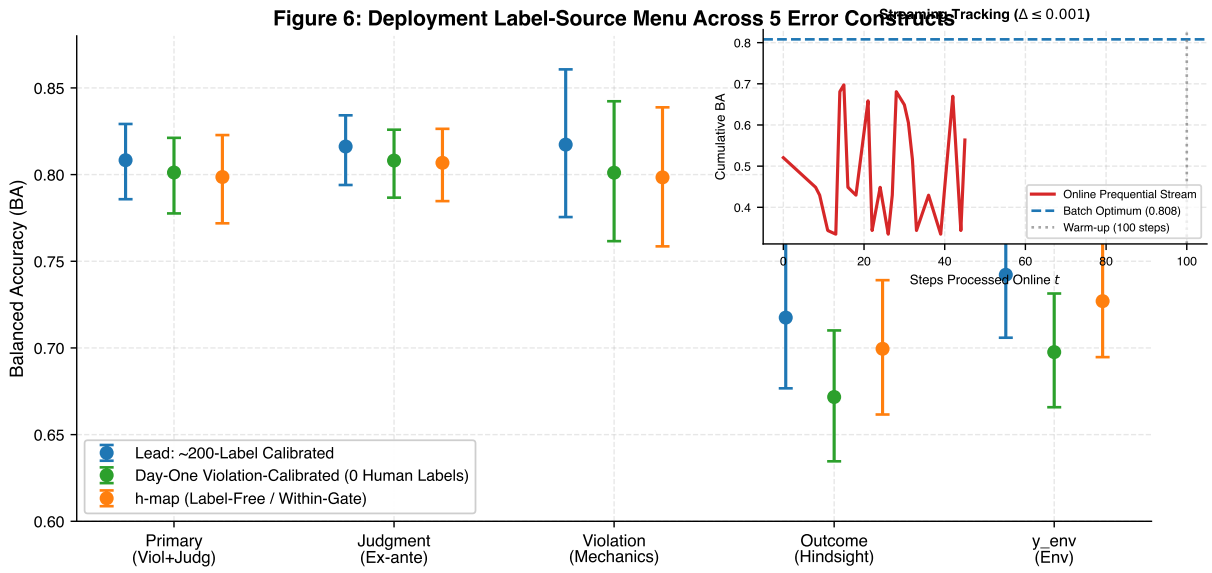


Figure 8: Three deployment routes at statistically indistinguishable primary-construct accuracy, with construct-robustness differences shown; inset, streaming operation converging to the batch rule. (The "∼200-label" route label is the fitted cut of Table 5.)

tested estimators partition identically under any bar inside a 0.20-wide gap (0.898/0.876/ 0.859 vs 0.679/0.590/0.546), and floors are evaluated on the primary construct only (an outcome-construct floor would qualify estimators the primary rejects: 0.759/0.744 on outcome for two that fail the primary). Two *non*-floors, with reasons: no floor on normalized separation, whose capable/weak ordering *inverted* under environment-derived labels (an ensemble-label artifact proven on identical steps); and no frontier-status shortcut, since gpt-4o sits below both capable open estimators. *Stability*: no fit-quality statistic we test — including one frozen in advance for a prospective test — predicts which placements miss; such statistics measure how well the map is determined, not how wrong it will be (full record in appendix). We retain the registered condition only as a conservative filter. *Scope*: external-only; per environment family (the competence map is an environment-family object: per-environment fits at $|r| = 0.948$–0.987 degrade to 0.717 pooled across environments); interpolation only within the reference pool's capability range. The claim: *the operating point is a characterizable, transferable property of the (estimator, environment) pair, with fit stability as a candidate validity diagnostic*: calibrate a threshold (Han et al., 2024) versus characterize an instrument.

7 Limitations

(i) The judgment-construct annotations are automated; they are validated at ∼2% error against environment-certain ground truth and treated as ex-ante ground truth throughout, with human agreement on the contested band untested. (ii) Characterization is per environment family; family granularity is untested, and the cross-environment observation is n=2. (iii) Label-free placement is interpolation-only. Its stability condition has now been tested prospectively on two further estimators and failed both times, as did a replacement diagnostic frozen in advance; we report label-free placement as effective in aggregate (12 of 16 evaluable held-out cells across four estimators land inside the budget, and every cell, landed or missed, stays within 0.023 balanced accuracy of its own oracle cut) while no available diagnostic predicts which cells will not. (iv) Analyses computed on annotation labels are marked where they appear. (v) Two environments, one scaffold family. (vi) Uncertainty *dynamics* (how estimates should drive information-seeking) is deferred with the control loop; our per-step risk signal is the primitive such models consume.

8 Conclusion

For practitioners, the recipe is now short: pick one capable open estimator, verify it clears the discrimination floor on a labelled slice, read every agent's steps with it (including the agents you cannot open), keep the probability rather than the verdict, and set the cut with a few hundred labelled steps, with free violation signals, or with the transferred map. The cost is a rounding error next to an API judge that also reads worse. For the field, three defaults should change: intrinsic uncertainty should not be the default estimator, trajectory-level AUROC should not be the default evaluation, and no agent-UQ result should ship without naming its error construct, because the construct choice alone can flip the verdict. The instrument's remaining open problem is stated exactly: label-free placement works in aggregate across four estimator families, and predicting its rare misses in advance is unsolved; the diagnostics that fail measure how well the map is determined, not how wrong it will be. A standardized per-step risk unit across heterogeneous agents is the prerequisite for the next problems (intervention policies, and failure attribution in multi-agent systems (Chen et al., 2026)), and the corpus, taxonomy, and characterization protocol released here are built to start them.

Reproducibility statement

Seed 13 throughout; bootstrap 2,000 draws; prequential seeds 13–22; evaluation scripts and configurations released with checksums; every headline number keyed to its source artifact in a released numbers registry; both label passes and all five per-step label columns released with provenance at `https://anonymous.4open.science/r/DACS-AUQ-7B29/`.

Ethics statement

No human-subjects data; trajectories are synthetic interactions with simulated environments. Judge-derived labels are disclosed as such rather than presented as human ground truth. Cost measurements are scoped to one tested frontier model and one task family and are released to inform deployment choices, not to characterize providers.

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A Result tables and figures

A.1 Per-Cell Evaluation Tables (Both Label Passes)

Independence Deltas (Table T2). Table 4 details the full 11-arm independence evaluation across all five error constructs.

Table 4: Full per-cell independence comparison: Target self-assessment vs capable external estimator.

Construct	Dataset	Target Model	Self AUROC	Ext AUROC	Δ AUROC	95% CI
Primary	ALFWorld	Llama-3.3-70B	0.860	0.897	+0.038	[+0.022, +0.054]
Primary	ALFWorld	Mistral-7B	0.634	0.882	+0.248	[+0.221, +0.273]
Primary	ALFWorld	Phi-4-mini	0.463	0.853	+0.390	[+0.362, +0.418]
Primary	ALFWorld	Qwen-35B	0.691	0.762	+0.071	[+0.047, +0.096]
Primary	ALFWorld	DeepSeek-V4	0.721	0.857	+0.129	[+0.096, +0.163]
Primary	ALFWorld	Gemma-4B	0.477	0.912	+0.436	[+0.400, +0.471]
Primary	HotpotQA	Llama-3.3-70B	0.902	0.893	-0.009	[-0.026, +0.007]
Primary	HotpotQA	Mistral-7B	0.706	0.913	+0.207	[+0.187, +0.228]
Primary	HotpotQA	Phi-4-mini	0.557	0.944	+0.387	[+0.363, +0.410]
Primary	HotpotQA	Qwen-35B	0.921	0.903	-0.018	[-0.034, -0.002]
Primary	HotpotQA	Gemma-4B	0.751	0.949	+0.198	[+0.177, +0.218]

B Prompts and reproducibility

B.1 Prompt Templates and Reproducibility Specifications

Prompt Template for Online External Estimation ($P(\textbf{True})$).

```
You are evaluating the correctness of an AI agent's proposed step in context.
[Task Description]: {task}
[Interaction History]: {history}
[Agent Reasoning]: {thought}
[Proposed Action]: {action}
Is the proposed step correct, productive, and admissible? Answer 'Yes' or 'No'.
Answer:
```

The trust score U_t is extracted via the normalized logit mass:

$$U_t = \frac{\exp(z_{\text{No}})}{\exp(z_{\text{Yes}}) + \exp(z_{\text{No}})} \tag{1}$$

Hardware Throughput and Sizing Constants.

- **Online Prefix Length**: 3,300 assessments / 5 minutes / A100 GPU (KV-prefix shared).
- **Hindsight Full-Length**: 375 assessments / 5 minutes / A100 (an $8.8\times$ compute expansion factor).
- **Fixed Seeds**: Master seed 13; Prequential seeds 13–22; Bootstrap draws 2,000.

C Methods in full

Every procedure the experiments compare against is specified here, so the paper stands without the repository. Each subsection names the file it was run from and that file's SHA-256 at the time of the runs reported.

Table 5: What each route actually annotates. The calibrated route fits its cut on every labelled step in half of the target's episodes, so its annotation cost is a property of the cell, not a fixed budget; the learned critic's budgets are fixed step counts and are reported as the number of episodes those steps span. Median and range are over the 22 capable cells of the deployment table.

route	annotated steps	annotated episodes	annotation units fixed?
calibrated cut	1436 (894–3254)	70 (60–250)	no
day-one violation cut	0	0	yes (none)
learned critic, 200 labels	200	12 (5–56)	yes
learned critic, 1,000 labels	1,000	42 (22–220)	yes
transferred map	0	0	yes (none)

Table 6: Held-out placement in accuracy terms. *Regret* is the balanced accuracy given up by cutting at the transferred percentile rather than at the cell's own optimum; intervals are episode-clustered bootstraps on the paired difference. The four cells that miss the ± 8-point tolerance give up no more accuracy than the twelve that land.

estimator	env.	miss (pts)	BA transferred	BA oracle	regret	95% CI
Llama-3.3-70B	ALFWorld / Qwen3.5-4B	0.04	0.6963	0.6963	0.0000	[0.0000, 0.0059]
Llama-3.3-70B	ALFWorld / Qwen3.5-9B	6.10	0.6829	0.6876	0.0047	[0.0018, 0.0170]
Llama-3.3-70B	HotpotQA / Qwen3.5-4B	2.79	0.8164	0.8389	0.0225	[0.0093, 0.0398]
Llama-3.3-70B	HotpotQA / Qwen3.5-9B	3.23	0.8295	0.8415	0.0120	[0.0041, 0.0330]
Qwen3.6-35B	ALFWorld / Qwen3.5-4B	7.33	0.6958	0.7003	0.0044	[0.0016, 0.0176]
Qwen3.6-35B	ALFWorld / Qwen3.5-9B	11.48	0.7018	0.7166	0.0148	[0.0067, 0.0323]
Qwen3.6-35B	HotpotQA / Qwen3.5-4B	10.15	0.8357	0.8420	0.0063	[0.0024, 0.0203]
Qwen3.6-35B	HotpotQA / Qwen3.5-9B	4.70	0.8206	0.8294	0.0089	[0.0053, 0.0241]
Gemma-3-27B	ALFWorld / Qwen3.5-4B	15.52	0.6170	0.6245	0.0075	[0.0014, 0.0201]
Gemma-3-27B	ALFWorld / Qwen3.5-9B	7.51	0.6528	0.6622	0.0094	[0.0013, 0.0213]
Gemma-3-27B	HotpotQA / Qwen3.5-4B	1.96	0.8062	0.8270	0.0207	[0.0094, 0.0372]
Gemma-3-27B	HotpotQA / Qwen3.5-9B	0.93	0.7962	0.8097	0.0134	[0.0036, 0.0315]
Mistral-Small-24B	ALFWorld / Qwen3.5-4B	8.39	0.6061	0.6155	0.0094	[0.0046, 0.0242]
Mistral-Small-24B	ALFWorld / Qwen3.5-9B	1.30	0.5927	0.6013	0.0086	[0.0050, 0.0200]
Mistral-Small-24B	HotpotQA / Qwen3.5-4B	2.01	0.7857	0.7943	0.0086	[0.0022, 0.0247]
Mistral-Small-24B	HotpotQA / Qwen3.5-9B	0.21	0.7789	0.7937	0.0148	[0.0043, 0.0347]
all evaluable cells		5.23			0.0104	[0.0077, 0.0132]
in range (reference score band)		5.10			0.0106	[0.0075, 0.0138]
out of range (reference score band)		6.11			0.0091	[0.0089, 0.0094]
inside the tolerance budget		3.18			0.0107	[0.0072, 0.0143]
outside the tolerance budget		11.38			0.0095	[0.0069, 0.0130]

C.1 Intrinsic self-assessment: metrics and evidence scopes

Source: `src/probes.py`, sha256 `0c99cac7240ec9153c5417756b2cf6d8d35e0c647ae6253649250162a5d1f7cd`.

Every intrinsic reading is a second call to the same model that produced the step, given the agent's own pre-action context plus the step's reasoning and proposed action. Three metrics were run; all three are reported as *uncertainty* U, with 1 meaning maximally uncertain, so higher always means more likely incorrect.

$P(\textbf{True})$**.** The model is asked a closed question — "Is the proposed action above the correct and appropriate next action for this task? Answer with a single word: Yes or No." — and the answer is read from the *first token's* top-logprobs rather than from generated text. Probability mass is summed over case and leading-space variants of Yes and of No, and $U = P(\text{No})/(P(\text{Yes}) + P(\text{No}))$. A record with no yes/no mass in the head is marked unparsed and excluded; it is never imputed. This is the only metric that needs logprob access, which is why a frontier model that refuses logprobs can be the agent in this framework but not the estimator.

Table 7: Self-leniency with and without model-generated ground truth. The *violation* column restricts the labels to environment rule violations, so no annotation model appears anywhere in the loop. Depression is the mean over 11 arms of (external score − self score) on the same steps; the competence rows give the self-cell's error-rate estimate against its truth.

measurement	annotation	primary	violation only
score depression, incorrect steps	$+0.112$	$+0.114$	$+0.102$
score depression, correct steps	$+0.017$	$+0.016$	$+0.052$
(arms with the effect in the expected direction)	7/11	7/11	6/11
competence error, self-cell (ALFWorld / Qwen3.6-35B)	-0.135	-0.170	-0.121
competence error, self-cell (ALFWorld / Llama-3.3-70B)	-0.006	$+0.002$	$+0.006$
competence error, self-cell (HotpotQA / Qwen3.6-35B)	-0.082	-0.070	$+0.034$
competence error, self-cell (HotpotQA / Llama-3.3-70B)	$+0.007$	-0.001	-0.050
cross-cell mean $\lvert\Delta p\rvert$ (comparison)	0.060	0.060	0.043
fits where the self-cell is the leverage point	2/4	1/4	2/4

Table 8: Evidence configuration behind every headline result. All headline numbers use the same bundle: the estimator is given exactly the context the agent had when it proposed the step, plus the step itself. An automated check of 28,050 sampled steps across every scoring pass confirmed the estimator's history block is identical to the agent's own pre-action history in 28,050 of them.

result	scoring unit	evidence given to the estimator	includes this step's result?
independence (self vs. external)	whole step, one probe	prior steps + this step's reasoning and proposed action	no
score-over-verdict channel	whole step, one probe	prior steps + this step's reasoning and proposed action	no
estimator tier	whole step, one probe	prior steps + this step's reasoning and proposed action	no
frontier comparison	whole step, one probe	prior steps + this step's reasoning and proposed action	no
operating-point map	whole step, one probe	prior steps + this step's reasoning and proposed action	no
deployment menu	whole step, one probe	prior steps + this step's reasoning and proposed action	no
deferral demonstration	whole step, one probe	prior steps + the proposed step, read before it is executed	no

Separate verbalized confidence. The model is asked, in a separate call, how confident it is that the step is correct, and answers in words. The reply is parsed to a number in $[0, 1]$ and $U = 1 - \text{confidence}$. This is the "grounded verbalized channel" the survey literature hopes for; on our measurements it is the inverted one for capable models.

Post-hoc numeric confidence. The same question on an explicit 0–100 scale ("On a scale of 0-100, how confident are you that the agent's response above …"), parsed to an integer and mapped to $U = 1 - n/100$. Unparseable replies are excluded on the same rule.

Probe scope. Every headline result uses a single whole-step probe: one call about the step as a unit, reasoning together with the chosen action. Narrower variants (reasoning-only, action-only, and their mean) exist in the release and underlie the recipe-stability analysis of §5.3; no headline number uses them. Probe seeds are derived per (step, probe, stage) so no two probe calls on a step share an RNG stream.

C.2 The supervised process critic

Source: `analysis/rev_critic.py`, sha256 `b9331ffd0001ad405712c1bbf659095f9a4288d8d355137bc2fa31404a`

A per-step binary classifier trained on cheap trajectory features, with no additional model calls, evaluated per capable external cell on the primary construct.

Features (leak-free set). Step index and $\log(1 + \text{index})$; episode length so far; action string length in characters and in words; a one-hot action type over the sixteen ALFWorld/HotpotQA verbs (`go, take, open, close, put, toggle, heat, cool, clean, examine, inventory, look, search, lookup, finish, none`);

Table 9: How much labelling the calibrated cut needs. Cells, episode split and evaluation half are identical across rows; only the number of labelled steps used to fit the cut changes. The cut saturates by roughly 500 labelled steps, within 0.004 of fitting on the whole half. The 1,000 and 2,000 rows cover fewer cells, because a cell's fitting half is often smaller than the budget, and are not comparable to the rest.

labelled steps used	cells	mean BA	95% CI
100	22	0.7843	[0.7556, 0.8101]
200	22	0.7889	[0.7599, 0.8140]
500	22	**0.7977**	[0.7735, 0.8197]
1,000	18	0.7956	[0.7683, 0.8207]
2,000	6	0.7917	[0.7789, 0.8066]
all of the fitting half	22	0.8014	[0.7780, 0.8220]

Table 10: Multiplicity-adjusted counts within each family of tests. Every test is a two-sided episode-clustered bootstrap on the paired difference (20,000 draws, seed 13); Holm–Bonferroni is applied within a family at a family-wise level of 0.05. *Direction* splits the surviving tests by the sign of the difference.

Family	Tested claim	Tests	$p \leq 0.05$	Holm	Direc
independence	an external reading beats the agent's own self-assessment (11 agents x 5 label constructs)	55	42	37	37 + /
channel	reading the score beats reading the estimator's own yes/no answer (60 estimator-agent cells)	60	46	40	40 + /
tier	a mid-scale estimator's calibrated cut beats a large estimator's answer (11 agents)	11	7	6	6 + /
frontier	a frontier judge differs from a capable open judge (22 cells, primary construct)	22	12	8	1 + /

plan length before the step, the change in plan length, and a plan-missing indicator; an environment indicator; and a one-hot over agent identity. The *hybrid* variant adds history-only summaries of the external estimator's score over steps strictly before t (mean, max, min, count, last value), which cannot leak U_t.

What is excluded, and why it must be. Five features are withheld from the leak-free set: `in_admissible`, `empty_result`, `loop_flag`, `repeat_dist`, `repeat_count`. Under the primary construct these are not predictors of the label — they are the Tier-A rules that *define* its positive class, so a critic given them recomputes the label rather than predicting it. Including them moves the comparison by more than 0.12 balanced accuracy and produces an apparent critic win; the leak-free set is the honest comparison and is the one reported.

Models and hyperparameters, fixed in advance. Logistic regression (`C`$= 1.0$, `max_iter`$= 2000$, lbfgs solver) on standardized features, and gradient-boosted trees (100 estimators, max depth 3, learning rate 0.1). No hyperparameter search was run at any point.

Budget protocol. Episodes are sorted by identifier and split in half; both fold directions are run and averaged. Within the fitting half, steps are taken *in episode order* — not sampled — until the budget is reached, giving budgets of exactly 200 and $1{,}000$ labelled steps. The estimator's own cut is fitted on the identical steps, so the two sides of the comparison see the same labels. Scoring is out-of-sample on the other half, and the $1{,}000$-step row covers only the cells whose fitting half contains that many labelled steps.

C.3 The unsupervised mixture cut

Scores are mapped to the logit scale and a two-component univariate Gaussian mixture is fitted by EM: k-means initialisation on perturbed quartiles, three restarts keeping the best log-likelihood, at most 200 iterations, convergence at 10^{-6}. The cut is the equal-posterior boundary between the two components, found on a 4001-point grid between the fitted means (the closed-form quadratic root can fall outside the data range), then mapped back through the logistic.

Abstention rule. The arm declines to place a cut, rather than placing a bad one, in four cases: EM failure; a minority component weight below 0.05; a separation between component means below 0.25 on the logit scale; or no sign change

Table 11: Recipe invariance with a third estimator. A *column* is one (estimator, environment, agent, construct) external cell, satisfied when the estimator's fixed evidence scope is within 0.02 AUROC of the best scope for that cell; a *cross arm* is one (estimator, environment, agent) cell on the primary construct, stable when the fixed scope is the best one there. All three estimators select the action-anchored scope.

Estimators	Columns	within 0.02	Cross arms	scope stable
Llama-3.3-70B, Qwen3.6-35B	90	88 (97.8%)	18	15 (83.3%)
+ Gemma-3-27B	145	133 (91.7%)	29	25 (86.2%)

Table 12: Label-source menu across error constructs and deployment routes. The registered primary construct is violation+judgment.

Construct	Deployment Route	Labels Required	Source	Mean BA
violation+judgment	day one – violation-calibrated	0	environment emits automatically	0.8012
	full labelling pass (saturates ∼500 steps) – fitted cut	1,436 med. steps	human or judge ensemble	0.8083
	steady state – h-rule	0	other targets of same judge x env	0.7986
judgment	day one – violation-calibrated	0	environment emits automatically	0.8081
	full labelling pass (saturates ∼500 steps) – fitted cut	1,436 med. steps	human or judge ensemble	0.8162
	steady state – h-rule	0	other targets of same judge x env	0.8068
violation	day one – violation-calibrated	0	environment emits automatically	0.8011
	full labelling pass (saturates ∼500 steps) – fitted cut	1,436 med. steps	human or judge ensemble	0.8173
	steady state – h-rule	0	other targets of same judge x env	0.7984
outcome	day one – violation-calibrated	0	environment emits automatically	0.6717
	full labelling pass (saturates ∼500 steps) – fitted cut	1,436 med. steps	human or judge ensemble	0.7175
	steady state – h-rule	0	other targets of same judge x env	0.6995
y_env	day one – violation-calibrated	0	environment emits automatically	0.6976
	full labelling pass (saturates ∼500 steps) – fitted cut	1,436 med. steps	human or judge ensemble	0.7422
	steady state – h-rule	0	other targets of same judge x env	0.7270

of the posterior difference between the means. Abstentions are counted as losses wherever a pass rate is reported — never dropped.

The procedure uses no labels at all, only the target's own score distribution.

C.4 The fixed-percentile and error-rate-indexed rules

Sources: `the released cut-transfer module`, sha256 `3021afcb3a716bd4a4985a77dc888daeb14334cf6f1ca90`
`the released indexed-cut module`, sha256 `9463a268b096b146f8429d7a111b39b311815a7398cafc1cd69356`

Both rules transfer from the same reference pool the operating-point map uses: all other targets of that (estimator, environment), self-cells excluded.

Raw threshold transfer. Pool the reference cells' steps, fit the balanced-accuracy-optimal cut on the pool, and apply that *score value* directly to the new target. This is the baseline the level-shift result rules out.

Fixed percentile. Take the same pooled cut, record what percentile it occupies within the pooled scores, and place the new target's cut at that percentile of its own unlabelled scores. Transfers a rank rather than a level, and needs no labels on the target.

Error-rate-indexed rule. Fit a linear map g from a reference cell's mean score to its error rate, over the same pool. For a new target, predict $\hat{p} = g(\bar{u})$, clipped to $[0.02, 0.98]$, and cut at the $100(1-\hat{p})$-th percentile of its own scores. This embeds the identity "optimal percentile $= 100(1-p)$", which the measurements contradict: realised optimal percentiles deviate from it by up to 26 points, so even an oracle handed the true error rate places the cut suboptimally.

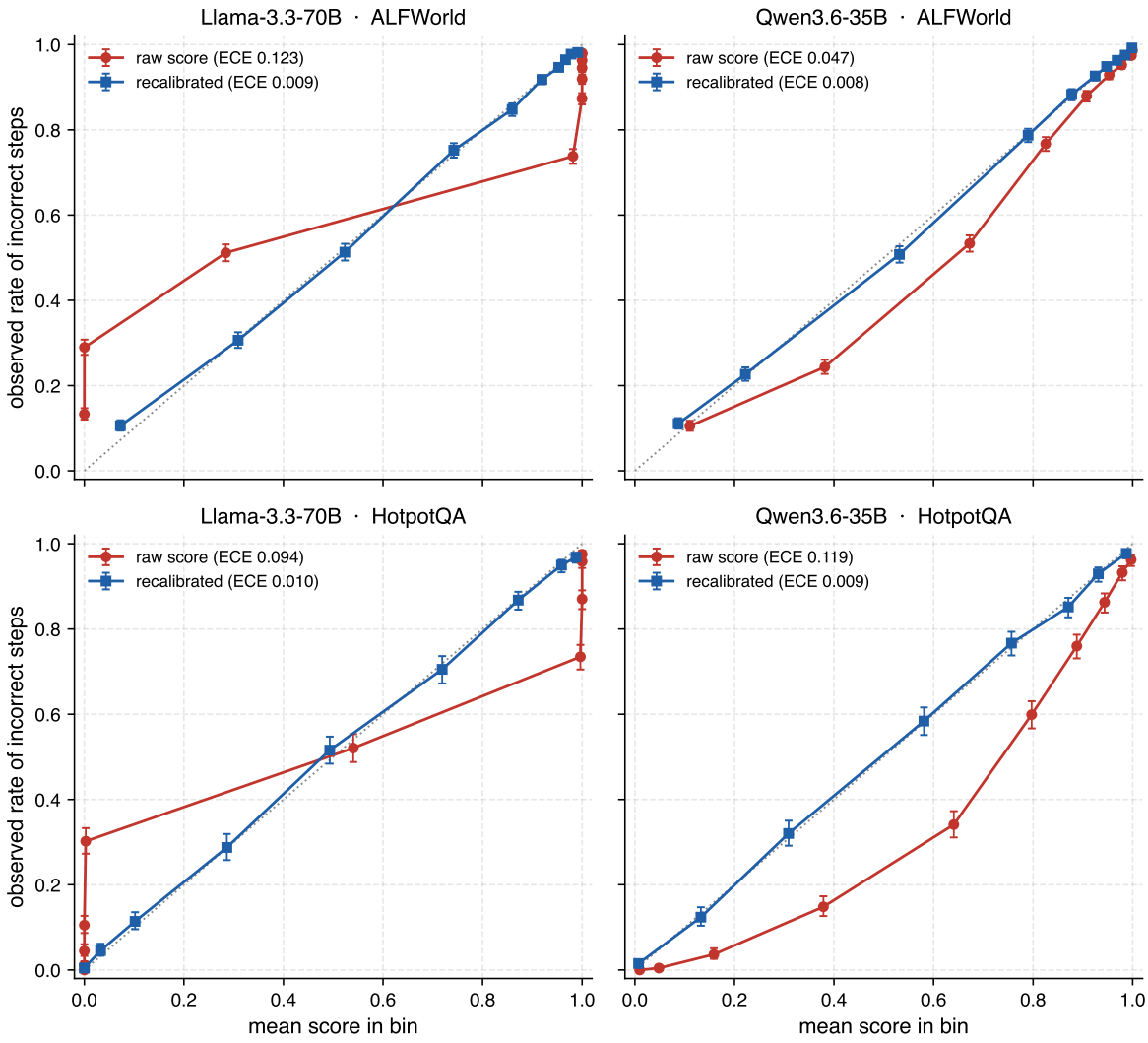


Figure 9: Reliability diagrams per (estimator, environment), raw and after isotonic recalibration fitted on the operating point's labelled episodes.

Common evaluation. All rules place a cut without reading any label of the target being scored, and all are evaluated by balanced accuracy against the same labels on the same steps. Realised flag rates match requested percentiles to within 0.001; no tie-lumping occurs.

D Probabilistic calibration of the risk score

Raw scores rank steps well but are not probabilistically calibrated: across the 18 cross-agent cells, expected calibration error averages 0.109 (range 0.011–0.287), with mass piling at both endpoints. A single monotone recalibration fitted on the half of the episodes already labelled for the operating point removes most of this (mean ECE 0.019; Brier $0.130 \to 0.106$; NLL $0.775 \to 0.358$) while leaving discrimination untouched: the logistic map changes balanced accuracy at the fitted cut by exactly zero in all cells, the isotonic map by at most 0.012. The raw score's Brier already beats the estimator's own binary answer in 18 of 18 cells and the constant predictor in 17 of 18.

Table 13: Probabilistic calibration of the per-step scores, primary construct. ECE is over ten equal-mass bins; Brier and ECE under *recalibrated* come from an isotonic map fitted on one half of the episodes and applied to the other, using the same episode split as the label-calibrated cut, so no additional annotation is spent. $\bar{y}$ is the cell's rate of incorrect steps; the constant column is the Brier score of predicting $\bar{y}$ for every step and the verdict column that of the estimator's own yes/no answer.

				ECE		Brier		Brier baselines	
Estimator	Trajectory source	n	$\bar{y}$	raw	recal.	raw	recal.	constant	verdict
ALFWorld									
Llama-3.3-70B	Llama-3.3-70B †	4285	0.53	0.193	0.022	0.210	0.155	0.249	0.228
Llama-3.3-70B	Mistral-7B	6203	0.80	0.121	0.012	0.130	0.103	0.160	0.143
Llama-3.3-70B	Phi-4-mini	6005	0.82	0.114	0.010	0.118	0.097	0.145	0.125
Llama-3.3-70B	Qwen3.6-35B	2872	0.42	0.287	0.055	0.299	0.201	0.244	0.314
Llama-3.3-70B	DeepSeek-V4-Flash	3067	0.34	0.174	0.031	0.201	0.154	0.224	0.219
Llama-3.3-70B	Gemma-3-4B	6507	0.91	0.038	0.008	0.059	0.051	0.082	0.063
Qwen3.6-35B	Llama-3.3-70B	4285	0.53	0.135	0.028	0.155	0.127	0.249	0.230
Qwen3.6-35B	Mistral-7B	6203	0.80	0.021	0.012	0.100	0.099	0.160	0.136
Qwen3.6-35B	Phi-4-mini	6005	0.82	0.057	0.008	0.097	0.092	0.145	0.128
Qwen3.6-35B	Qwen3.6-35B †	2872	0.42	0.149	0.035	0.243	0.224	0.244	0.366
Qwen3.6-35B	DeepSeek-V4-Flash	3067	0.34	0.104	0.039	0.149	0.142	0.224	0.212
Qwen3.6-35B	Gemma-3-4B	6507	0.91	0.011	0.005	0.051	0.052	0.082	0.064
HotpotQA									
Llama-3.3-70B	Llama-3.3-70B †	1878	0.18	0.090	0.019	0.120	0.095	0.149	0.130
Llama-3.3-70B	Mistral-7B	2626	0.56	0.143	0.017	0.151	0.117	0.246	0.166
Llama-3.3-70B	Phi-4-mini	2282	0.57	0.097	0.012	0.125	0.099	0.245	0.139
Llama-3.3-70B	Qwen3.6-35B	1787	0.11	0.069	0.018	0.088	0.069	0.095	0.095
Llama-3.3-70B	Gemma-3-4B	2248	0.48	0.114	0.029	0.124	0.100	0.250	0.136
Qwen3.6-35B	Llama-3.3-70B	1878	0.18	0.169	0.007	0.143	0.098	0.149	0.216
Qwen3.6-35B	Mistral-7B	2626	0.56	0.135	0.017	0.146	0.120	0.246	0.206
Qwen3.6-35B	Phi-4-mini	2282	0.57	0.084	0.014	0.104	0.090	0.245	0.148
Qwen3.6-35B	Qwen3.6-35B †	1787	0.11	0.110	0.015	0.086	0.066	0.095	0.121
Qwen3.6-35B	Gemma-3-4B	2248	0.48	0.096	0.013	0.106	0.090	0.250	0.153
pooled Llama-3.3-70B	ALFWorld (5 cells)	24654	0.73	0.123	0.009	0.137	0.106	—	—
pooled Qwen3.6-35B	ALFWorld (5 cells)	26067	0.73	0.047	0.008	0.102	0.095	—	—
pooled Llama-3.3-70B	HotpotQA (4 cells)	8943	0.45	0.094	0.010	0.125	0.098	—	—
pooled Qwen3.6-35B	HotpotQA (4 cells)	9034	0.46	0.119	0.009	0.125	0.100	—	—

† the estimator reading its own trajectories; pooled rows exclude those cells.

E Matched-capability independence analysis

Removing the family confound leaves the external advantage intact (same-family reading of the held-out agents: $+0.059$ mean AUROC, 4/4 cells, all intervals excluding zero). Removing the capability advantage removes most of it (peer-scale mean $+0.008$, external ahead in 1 of 4; sub-7B estimators reading one another: -0.074, ahead in 4 of 12, with 11 of 12 intervals excluding zero, most favouring self-assessment). The peer-scale asymmetry (one direction $+0.068$, the reverse -0.007 on the same environment) is the signature of self-leniency rather than reader quality. Reader capability dominates what external reading buys, with evidence of a smaller average self-leniency effect.

Figures

Table 14: What an external reading buys once the two confounds in the headline comparison are removed. Δ is the external reader's AUROC minus the agent's own best self-assessment, on identical steps, with a 95% episode-clustered bootstrap interval. The self baseline is the best of the agent's four reading scopes, chosen in hindsight, which can only favour it. Primary construct.

Environment	External reader	Agent read	n	external	self (best)	Δ [95% CI]
Same family: the family confound removed						
ALFWorld	Qwen3.6-35B	Qwen3.5-4B	3754	0.767	0.665	$+0.102$ [$+0.079$, $+0.127$]
HotpotQA	Qwen3.6-35B	Qwen3.5-4B	1914	0.922	0.877	$+0.045$ [$+0.031$, $+0.059$]
ALFWorld	Qwen3.6-35B	Qwen3.5-9B	3859	0.781	0.728	$+0.053$ [$+0.036$, $+0.071$]
HotpotQA	Qwen3.6-35B	Qwen3.5-9B	1202	0.911	0.876	$+0.035$ [$+0.018$, $+0.053$]
ALFWorld	Qwen3.6-35B	Qwen3.5-27B	*no usable paired cell (single class or ¡40 shared labelled steps)*			
HotpotQA	Qwen3.6-35B	Qwen3.5-27B	*no labelled steps for this agent*			
Matched scale: the larger-judge confound removed						
ALFWorld	Qwen3.6-35B	Llama-3.3-70B	4285	0.897	0.904	-0.007 [-0.024, $+0.010$]
ALFWorld	Llama-3.3-70B	Qwen3.6-35B	2872	0.762	0.694	$+0.068$ [$+0.036$, $+0.098$]
HotpotQA	Qwen3.6-35B	Llama-3.3-70B	1878	0.893	0.902	-0.009 [-0.026, $+0.008$]
HotpotQA	Llama-3.3-70B	Qwen3.6-35B	1787	0.903	0.921	-0.018 [-0.035, -0.002]
Small reading small: externality without capability						
ALFWorld	Mistral-7B	Phi-4-mini	6005	0.620	0.547	$+0.073$ [$+0.046$, $+0.099$]
ALFWorld	Mistral-7B	Gemma-3-4B	6507	0.730	0.668	$+0.062$ [$+0.029$, $+0.094$]
ALFWorld	Phi-4-mini	Mistral-7B	6203	0.468	0.646	-0.179 [-0.209, -0.148]
ALFWorld	Phi-4-mini	Gemma-3-4B	6507	0.514	0.668	-0.155 [-0.195, -0.114]
ALFWorld	Gemma-3-4B	Mistral-7B	6203	0.421	0.646	-0.226 [-0.257, -0.194]
ALFWorld	Gemma-3-4B	Phi-4-mini	6005	0.412	0.547	-0.136 [-0.163, -0.108]
HotpotQA	Mistral-7B	Phi-4-mini	2282	0.704	0.632	$+0.072$ [$+0.045$, $+0.097$]
HotpotQA	Mistral-7B	Gemma-3-4B	2248	0.727	0.751	-0.024 [-0.049, $+0.001$]
HotpotQA	Phi-4-mini	Mistral-7B	2626	0.540	0.769	-0.229 [-0.253, -0.205]
HotpotQA	Phi-4-mini	Gemma-3-4B	2248	0.595	0.751	-0.156 [-0.183, -0.129]
HotpotQA	Gemma-3-4B	Mistral-7B	2626	0.677	0.769	-0.093 [-0.116, -0.069]
HotpotQA	Gemma-3-4B	Phi-4-mini	2282	0.729	0.632	$+0.097$ [$+0.072$, $+0.122$]
Boundary: the cells where self-assessment is not beaten						
HotpotQA	Qwen3.6-35B	Llama-3.3-70B	1878	0.893	0.902	-0.009 [-0.026, $+0.008$]
HotpotQA	Llama-3.3-70B	Qwen3.6-35B	1787	0.903	0.921	-0.018 [-0.035, -0.002]

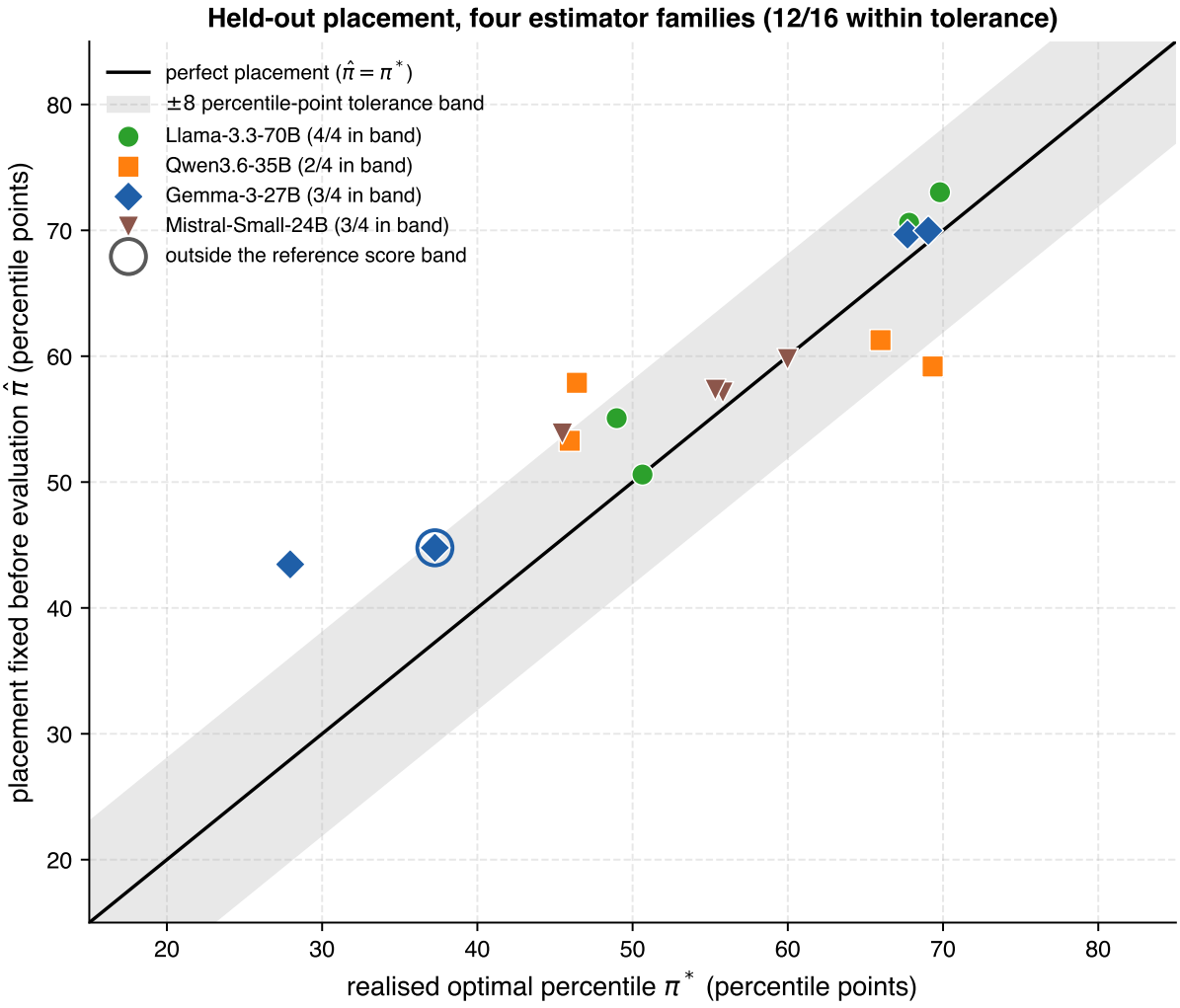


Figure 10: Held-out placement across four estimator families. Each point is one evaluable held-out agent: horizontal, the optimal cut percentile the labels turned out to require; vertical, the percentile fixed before any of those labels were read. The shaded band is the ±8-point tolerance; 12 of 16 cells fall inside it, zero labels spent on the target. Rings mark cells whose mean score lies outside the reference band, where the interpolation-only scope condition does not license deployment; the two later estimators were the weakest qualifiers, and their misses fall where reference scores compress.

Figure 9: Balanced Accuracy Flatness Curves Across Cut Percentiles (Median Flat Region: 16% Points)

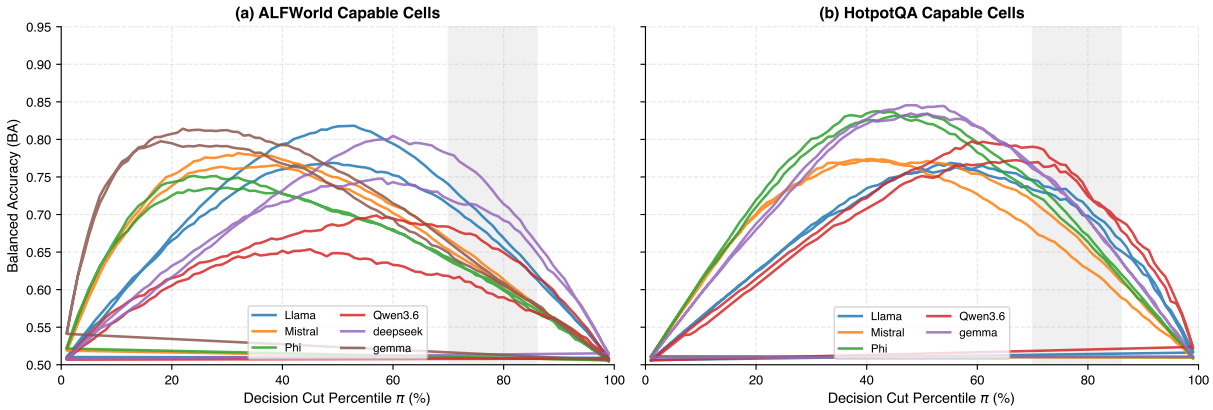


Figure 11: Balanced accuracy versus cut percentile per capable cell: the median flat region spans 16 percentile points, the error budget the operating-point map must meet.

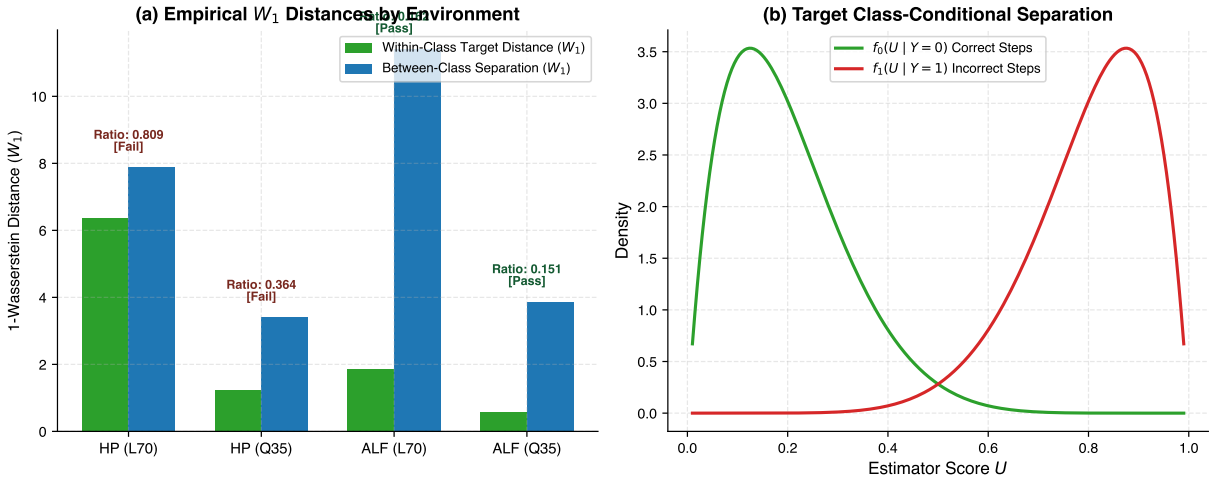


Figure 12: Class-conditional score distributions across targets: approximate invariance on HotpotQA, failure on ALFWorld, where the map nevertheless works; the mechanism is the flat landscape plus a coarse affine regularity.

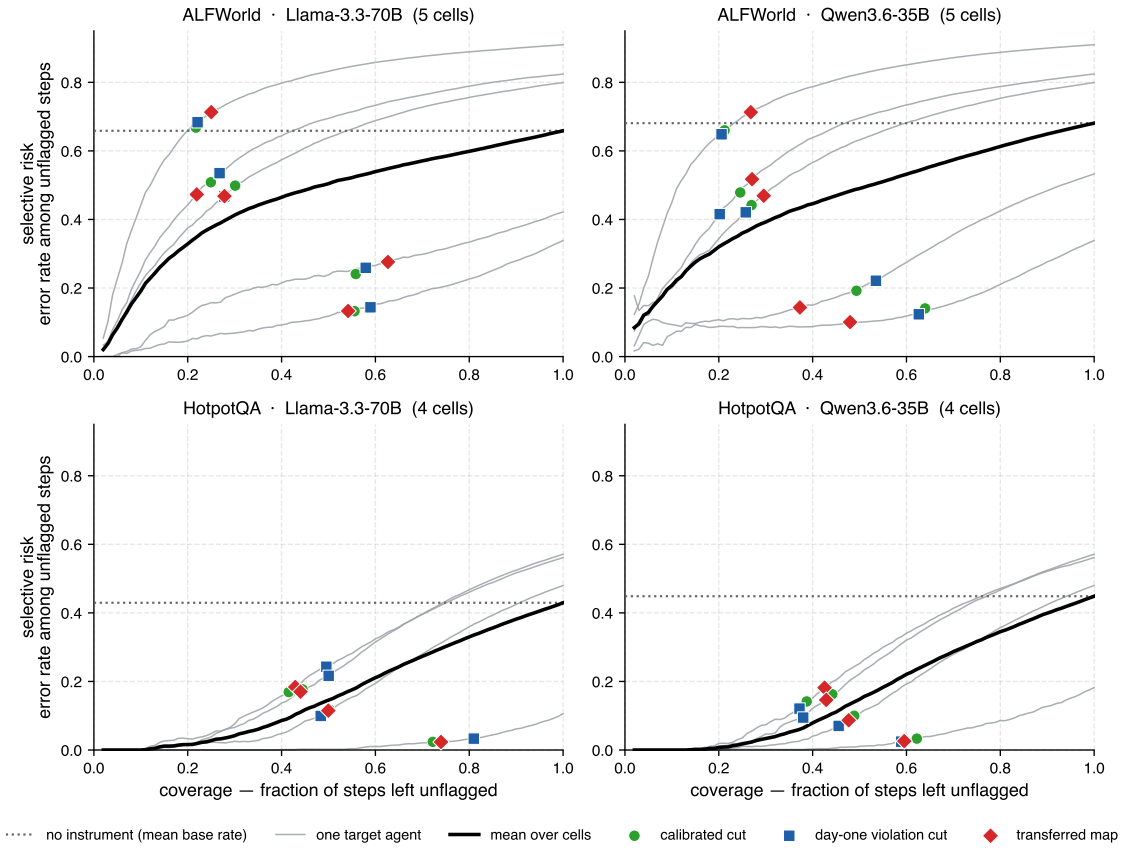


Figure 13: Selective risk versus coverage per (estimator, environment), with the three deployment routes' operating points marked; they coincide on the curve.